Zixu Wang

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EDUCATION

Undergraduate in Department of Automation August 2025 - Present
School of Information Science and Technology
Tsinghua University, GPA: 3.94/4.00

SELF-INTRODUCTION

Hi! I'm a highly motivated undergraduate student in the Department of Automation at Tsinghua University, with a strong passion for Embodied Intelligence, Physical AI and Multimodal Large Language Model. I'm eager to explore interdisciplinary research at the intersection of automation, computer science, and artificial intelligence, and to apply theoretical knowledge to real-world engineering challenges.

PERSONAL SKILLS

Embodied Intelligence	Algorithm Design (Reinforcement Learning, Imitation Learning, Motion Planning), Robot Simulation (Isaac Sim, Gazebo), ROS2, Sensor Fusion, Kinematics & Dynamics Modeling
Computer Vision	3D Vision (Point Cloud Processing, SLAM), 2D Vision (Object Detection, Segmentation, Tracking, Pose Estimation), Camera Calibration, Multi-View Geometry
Programming & Tools	Python, C/C++, Multisim, PyTorch

RESEARCH INTEREST

Embodied AI	Robot Learning (Reinforcement Learning, Imitation Learning, Sim-to-Real Transfer), Manipulation & Locomotion, Human-Robot Interaction, Physics Simulation & Digital Twins
Computer Vision	3D Scene Understanding, Neural Rendering (NeRF, 3D Gaussian Splatting), Visual SLAM & 3D Reconstruction, Multimodal Perception & Foundation Models
AI Foundations	Multi-Agent Systems, Sequence Modeling & Transformers, Representation Learning, Generative Models

PROJECTS

Critical Path Method Project Scheduler July 2026 (In Progress)
Object-Oriented Programming Training, Tsinghua University
- Developed a C++ project scheduling system implementing the Critical Path Method (CPM) algorithm
- Designed with MVC architecture and a template-based file I/O plugin system supporting extensible formats
- Supports four standard dependency types (FS, SS, FF, SF) with configurable lag, resource management, and DAG cycle detection via DFS-based topological check

Catgirl Resource Machine — A Programming Puzzle Game November 2025
Fundamentals of Programming, Tsinghua University
- Built a puzzle game in C++/Qt 6 inspired by *Human Resource Machine*, featuring an assembly-like instruction set (inbox, outbox, add, sub, copyto, copyfrom, jump, jumpifzero)
- Implemented a 4-level campaign with progressive difficulty, single-step debugging, and real-time visual feedback via Qt signal/slot mechanism
- Course grade: **100/100, A+, 4.0/4.0**; co-developed with one teammate

Home Service Robot Vision Pipeline (YOLO11-seg + SAM3) June 2026
Introduction to Artificial Intelligence, Tsinghua University

- Built a full-stack deep learning pipeline for household object detection and instance segmentation, targeting home service robotics applications
- Integrated SAM3 for zero-shot automatic annotation with seed-and-propagate workflow across video frames, and YOLO11-seg for real-time inference
- Implemented automatic dataset export (YOLO format), data augmentation (Albumentations), and a React + Vite web frontend with WebSocket-based real-time training feedback
- Course grade: **4.0/4.0**

ACADEMICAL EXPLORATIONS

Embodied Intelligent Service Robot Based on Multimodal Interaction for Open Scenarios,
College Student Innovation Training Program, Participant — February 2026

End-to-End VLA Development and Application for Dual-Arm Robots,
Academic Advancement Program, Participant — May 2026 – Present

End-to-End Incremental 7-DoF Robotic Grasping System Based on OpenVLA,
SRT (Student Research Training), Project Initiator — March 2026 – Present

POSITION OF RESPONSIBILITY

Hardware Department Officer, Student Association of Science and Technology,
Department of Automation, Tsinghua University — May 2026 – Present

Technical Commissioner, "General Artificial Intelligence" Class,
Department of Automation, Tsinghua University — May 2026 – Present

Vision Team Member, Tinker Team, Tsinghua Future Robot Club (FuRoC) — `tinkerfuroc.github.io`

EXTRA-CURRICULAR

- **Tsinghua 2nd Embodied Intelligence Challenge**, Team Member — December 2025
- **Tsinghua 5th Robot Dog Development Competition**, Team Leader — June 2026
- **2026 China Robot Competition & RoboCup China Open (RoboCup@Home)**, Team Member — 2026
- **RoboCup 2026 Incheon, RoboCup@Home League**, Team Member — July 2026

HONORS AND AWARDS

NOIP (National Olympiad in Informatics in Provinces) — Provincial First Prize — 2023

Tsinghua 2nd Embodied Intelligence Challenge — Special Award — December 2025

Tsinghua 5th Robot Dog Development Competition — 4th Place & Best Technological Innovation Award — June 2026

2026 China Robot Competition & RoboCup China Open (RoboCup@Home) — National First Prize — 2026

RoboCup 2026 Incheon, RoboCup@Home League — 11th Place Worldwide — July 2026

DECLARATION

I hereby declare that all the details furnished above are true to the best of my knowledge and belief.